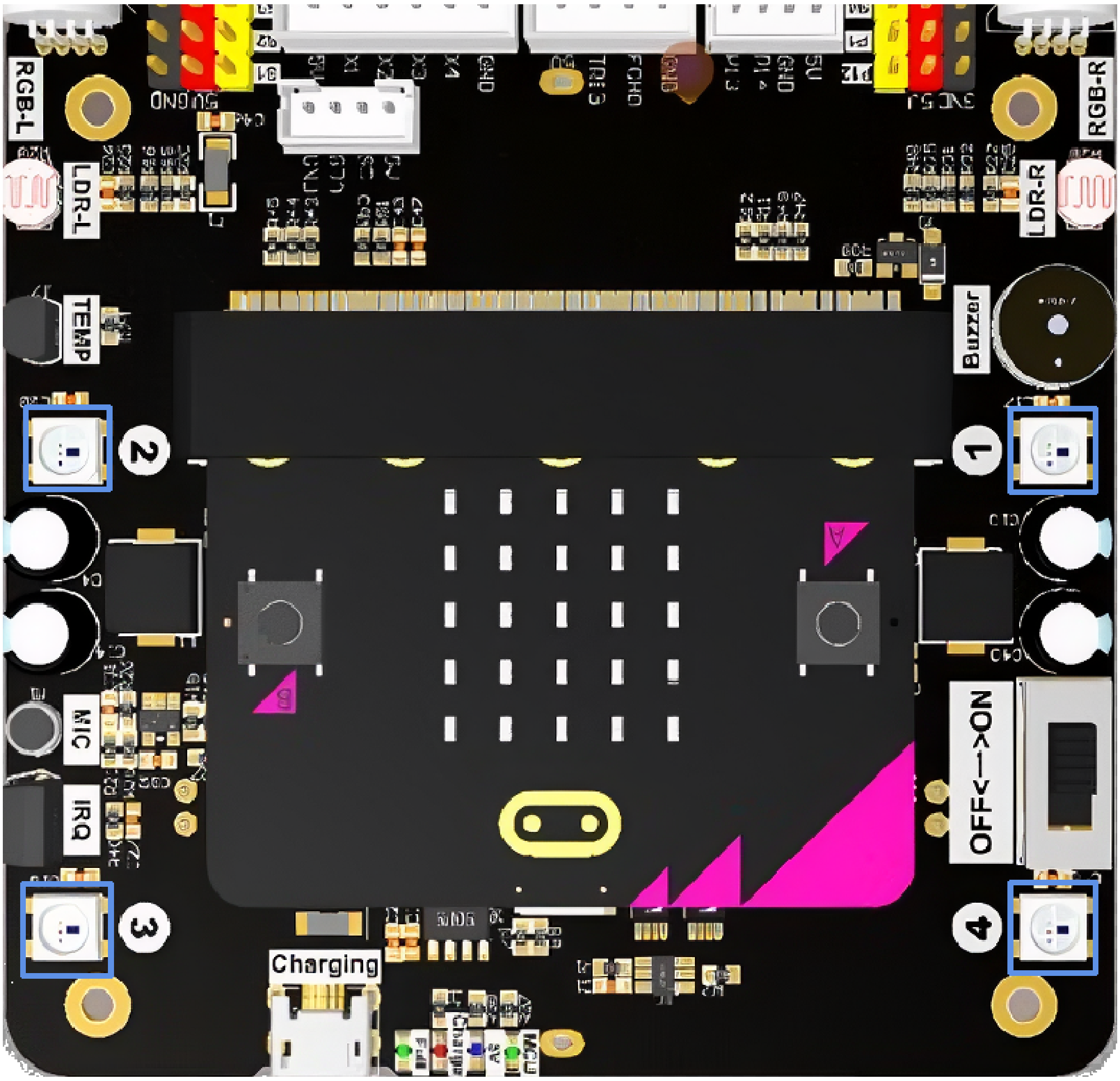
Light Up Colorful Lights

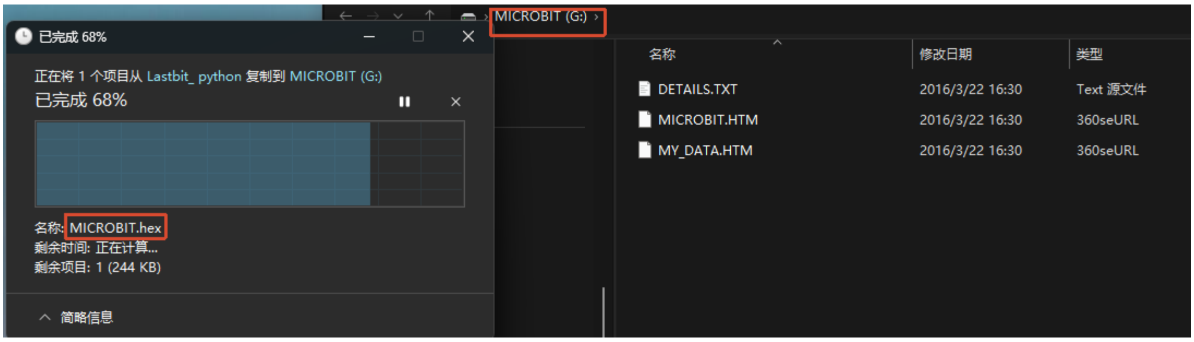
Feature Description

The rear of the Lastbit car is equipped with four RGB colorful lights. In this section, we will learn how to drive and light them up.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

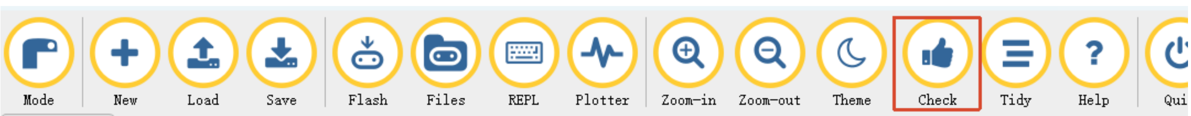
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

   Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.

Code Implementation

```python
# 3.Light_Up_Colorful_Lights
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

#  close all lights
Lastbit.all_lights_off()

while True:
    #  red light
    for i in range(4):
        Lastbit.set_ws_pixel(i, 'RED')
        # # Also supports passing in color tuples (R, G, B)
        # Lastbit.set_ws_pixel(i, (255, 0, 0))
    sleep(2000)

    #  green light
    Lastbit.set_ws_all('GREEN')
    # # Also supports passing in color tuples (R, G, B)
    # Lastbit.set_ws_all((0, 255, 0))
    sleep(2000)

    #  blue light
    for i in range(4):
        Lastbit.set_ws_pixel(i, 'BLUE')
        # Also supports passing in color tuples (R, G, B)
        # Lastbit.set_ws_pixel(i, (0, 0, 255))
    sleep(2000)

    #  close all seven color lights
    Lastbit.off_ws2812()
    sleep(2000)
```

`set_ws_pixel(i, color)`: sets a single colorful light. The first parameter `i` indicates which colorful light to set, and can be 0-3. The second parameter `color` can be one of the 9 colors, and it also supports passing in a color tuple (R, G, B).

- Red: 'RED'
- Orange: 'ORANGE'
- Yellow: 'YELLOW'
- Green: 'GREEN'
- Cyan: 'CYAN'
- Blue: 'BLUE'
- Purple: 'PURPLE'
- White: 'WHITE'
- Black (off): 'CLOSE'

`set_ws_all(color)`: supports setting the color of all colorful lights. The `color` can be one of the 9 colors, and it also supports passing in a color tuple (R, G, B).

`off_ws2812()`: turns off the colorful lights.

`all_lights_off()`: turns off all lights, including the body colorful lights and the front RGB spotlights.

Experimental Phenomenon

After the program is flashed, the colorful lights at the rear of the car will light up in a cycle of red -> green -> blue, switching the light effect every 2 seconds.